

Lot traveler and process source

Aster Microdevices | Fab 8 | AMX-4100 LVT gate flow

MRB HOLD

PROCESS TRAVELER

Step	Tool	Recipe	Start	End	Recorded state
Implant P3	IMP-17	BORON_LVT_042	05:12	05:58	Accepted
Ash	ASH-04	ASH_STD_18	06:10	06:34	Accepted
Wet clean	WET-12	SC1_SC2_09	06:48	07:22	Accepted
PVD TiN	PVD-03	TIN_GATE_31	08:06	09:44	Alarm 09:18
Anneal	RTP-08	RTP_LVT_22	10:01	10:29	Accepted
Metrology	CDSEM-06	GATE_LVT_QC	10:45	11:56	Reviewed

RECIPE SOURCE RECONCILIATION

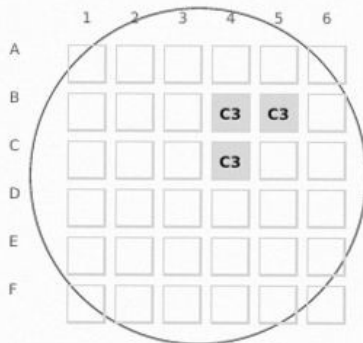
Source	Recipe shown	State	Use for this lot
Traveler page	TIN_GATE_31	Current	Yes
PVD alarm printout	TIN_GATE_30	Stale cached header	No
MES audit event	TIN_GATE_31	Confirmed	Yes / controls
Engineer note	Increase clamp purge	Future revision	Not used

MES event 88419 confirms TIN_GATE_31 executed on PVD-03. The TIN_GATE_30 string is confined to a stale alarm-printout header.

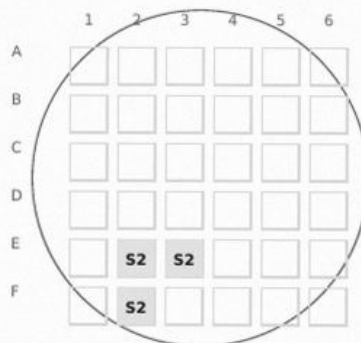
Wafer-map review

Grid A-F × 1-6 | reviewed defect codes shown in-map

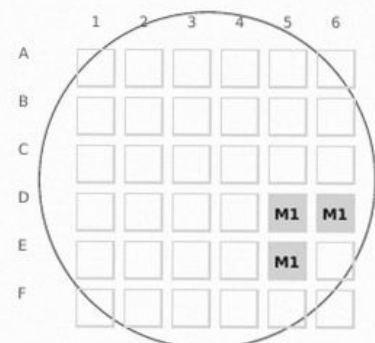
WAFER 03



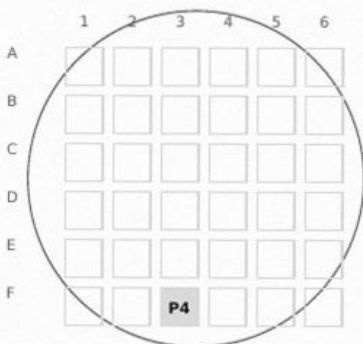
WAFER 05



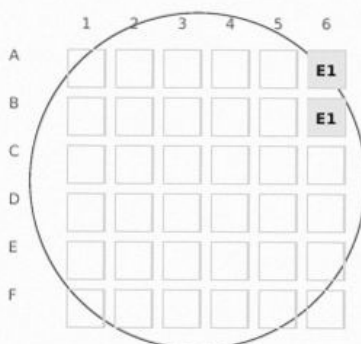
WAFER 07



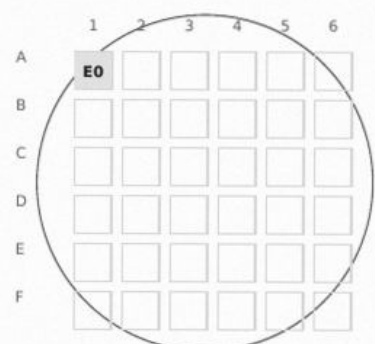
WAFER 09



WAFER 12



WAFER 02



Code	Class	Map color	Disposition gate
M1	Metal flake	Red	Scrap cluster
S2	Scratch	Orange	Engineering hold
E1	Edge bead	Yellow	Engineering hold
P4	Parametric drift	Blue	Reliability condition
C3	Center CD cluster	Violet	Remeasure / MRB
E0	Edge mark	Gray	Reference only

CD-SEM metrology review

Target 24.9 nm | action limits -2.0 / +2.5 nm

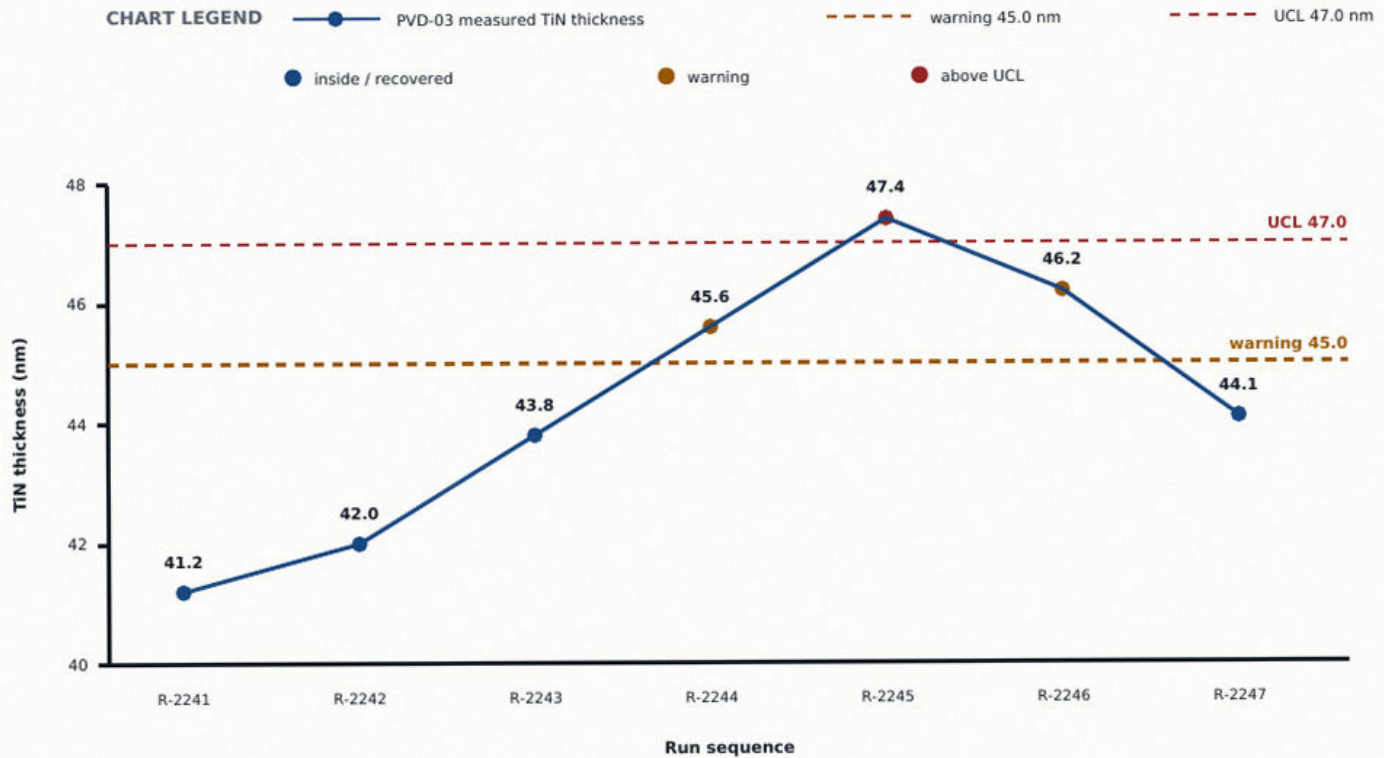
Wafer / site	CD nm	Delta nm	Flag	Image ref	Reviewer note
03 / B4	27.8	+2.9	H	IMG-221	Matches C3 map
03 / C4	27.4	+2.5	H	IMG-222	Adjacent high
05 / E2	24.1	-0.8	L-map	IMG-223	Scratch path
07 / D5	29.6	+4.7	H	IMG-224	Flake shadow
07 / D6	29.1	+4.2	H	IMG-225	Adjacent flake
09 / F3	26.8	+1.9	Watch	IMG-226	Parametric only
12 / A6	22.9	-2.0	L	IMG-227	Edge bead

DISPOSITION RULES

Rule	Condition	Required action
CD-H	Delta >= +2.5 nm	MRB disposition
CD-L	Delta <= -2.0 nm	MRB disposition
Map correlation	Defect code at measured site	Review image and map together
Yield exception	Wafer yield >93%	Does not waive action-limit review

PVD thickness SPC and alarm interlocks

PVD-03 sequence R-2241 through R-2247

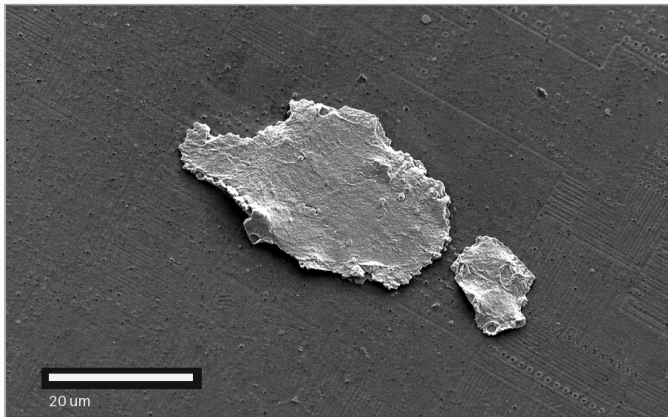


ALARM INTERLOCKS

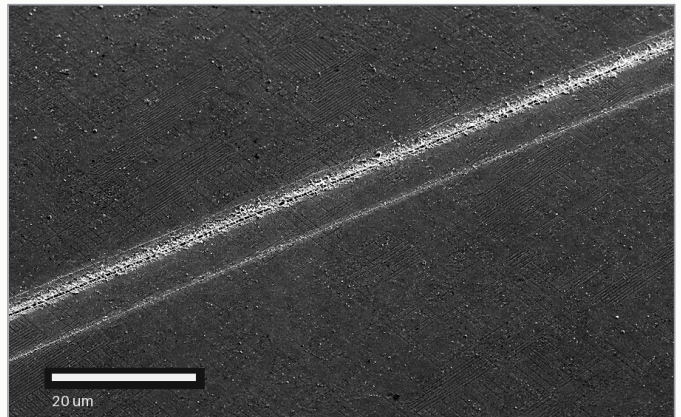
Interlock	Observed	Threshold	State	Disposition
Chamber pressure	4.8 mTorr	≤ 5.0	Pass	Not root cause
Clamp purge	18.1 slm	≥ 19.0	Fail	Future increase only
Ti target age	81.4 kWh	≤ 85	Watch	Clean then continue
Monitor residual	+1.7 nm	± 1.0	Fail	R-2245 review
Alarm ack	09:18 / J. Kwon	Required	Complete	Containment starts

SEM review contact sheet

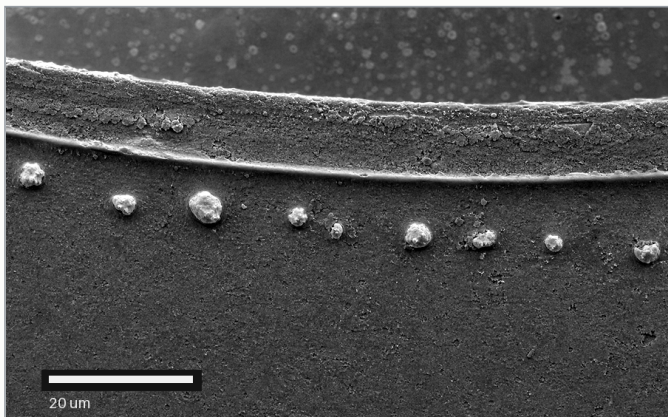
Representative reviewed frames; 20 um scale bars



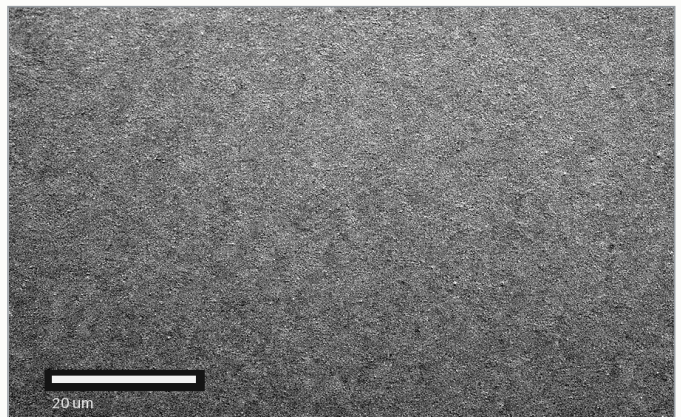
IMG-224 / W07 D5



IMG-223 / W05 E2



IMG-227 / W12 A6



IMG-226 / W09 F3

Defect classification and MRB decisions

Map, metrology, and image evidence reconciled at 11 June meeting

Code	Class	Count	Wafer	Gate
M1	Metal flake	44	07	Scrap
S2	Scratch	31	05	Engineering hold
E1	Edge bead	27	12	Engineering hold
P4	Parametric drift	19	09	Conditional
C3	Center CD cluster	16	03	Remeasure / hold

MATERIAL REVIEW BOARD

Item	Decision	Reason	Owner
Wafer 07	Scrap	Map W07-M1 plus IMG-224/225	Y. Nishida
Wafer 05	Hold	Map W05-S2 plus IMG-223	A. Roy
Wafer 03	Hold	Map W03-C3 plus metrology	M. Alvarez
Wafer 12	Hold	Map W12-E1 plus IMG-227	K. Stone
Wafer 09	Conditional	P4 parametric; REL-22-A required	S. Han
01,02,04,06,08,10,11	Release	Inside disposition limits	MRB

Reliability conditions and shipping allocation

Shipment authority remains conditional for wafer 09

Sample	Wafer	Stress	Result	Disposition use
REL-22-A	09	HTOL 168 h	Pending	Condition for shipment
REL-22-B	02	TC 100 cycles	Pass	Reference
REL-22-C	11	HAST 96 h	Pass	Reference
REL-22-D	07	Not built	Scrap	No reliability credit

SHIPPING ALLOCATION

Bucket	Wafers	Good die	Shipment condition
Release now	01,02,04,06,08,10,11	127,759	Standard COA
Conditional	09	17,880	Wait for REL-22-A pass
Engineering hold	03,05,12	52,144	Do not ship
Scrap	07	16,902	Exclude from yield and COA

HTOL CHAMBER LOAD / CONTROLLED WORK TICKET

Sample REL-22-A Wafer 09 Stress HTOL 168 h
Chamber H-17 Tray position C4 Loaded 11 Jun 14:22
Door seal witness KT Humidity logger HL-88 attached
Interim 24 h observation: no electrical reject
Final 168 h result: PENDING - wafer 09 remains segregated
Custody: S Han 11 Jun 14:22 / Quality M Alvarez 11 Jun 14:31



Defect-image verification worksheet

Inspector marks retained with the image archive routing record

ASTER MICRODEVICES / DEFECT IMAGE VERIFICATION

Lot Q8R7-22 Worksheet IQC-071 Review set IMG-223/224/226/227

[X] Wafer map compared with image archive

[X] Neighboring-die review completed for W07 cluster

[] Independent morphology confirmation completed

[] Preliminary all-wafer COA released

[N/A] Reticle excursion review (disabled for PVD-origin event)

~~IMG-224 Initial count: 1 flake~~

Corrected count: 2 reportable objects; see controlled image archive

Correction basis: second object resolved at archive magnification

Inspector R Kim 11 Jun 09:42 Reviewer A Roy 11 Jun 11:06

ARCHIVE ROUTING

Archive object	Custodian	Checksum state	Packet role
IQC-071	Incoming quality	Signed	Inspection worksheet
IMG set 223/224/226/227	Fab image archive	Matched	Visual evidence
MAP-11JUN	Yield engineering	Matched	Coordinate source
COA-DRAFT-02	Quality systems	Void	Not shipment authority

Certificate holdback and final audit trail

Signed quality record controls shipment eligibility

PARTIAL RELEASE

COA field	Signed value	Application
Lot	Q8R7-22	All wafers
Released wafers	01,02,04,06,08,10,11	Ship now
Conditional wafer	09	After REL-22-A pass
Engineering hold	03,05,12	Excluded
Scrap	07	Excluded from yield and COA
Executed recipe	TIN_GATE_31	MES confirmed

AUDIT TRAIL

Time	Actor	Entry	Effect
10 Jun 12:18	MES	PVD alarm attached	Containment starts
10 Jun 14:40	Yield engineering	Map review complete	03/05/07/12 flagged
11 Jun 08:55	MRB	07 scrap; 09 conditional	Disposition controls
11 Jun 10:20	Quality	COA holdback signed	Draft all-wafer COA void
11 Jun 13:30	Planning	Ship buckets created	127,759 die release

The preliminary all-wafer COA is void. Only the signed values above authorize shipment.